Team Problems

1. Compute the smallest positive integer for which the product of its digits is 96.

2. Compute the smallest integer $n > 1$ such that $10n$ has exactly 21 more factors than n.

3. The area of a right triangle ABC is 6 and the length of the hypotenuse $\overline{AB}$ is 10. Compute $\sin 2A$.

4. Let T be the set of all two digit positive integers, and let S_k be the sum of all elements of T *except* k. Compute the value of k such that S_k is a palindrome (that is, the digits of the number read the same forwards as backwards, such as 12321).

5. Let A, B, C, D, and E be a sequence of increasing positive integers. The set

$$\{18, 20, 21, 22, 23, 24, 25, 26, 28, 29\}$$

   is the set of all pairwise sums of distinct integers in the sequence. Compute the ordered 5-tuple (A, B, C, D, E).

6. Compute the surface area of the boundary of the following region:

$$\big\{(x, y, z) : x^2 + y^2 \leq 4, |x| \leq 1, |z| \leq 4\big\}.$$

7. Compute the number of ordered 4-tuples (a, b, c, d) of positive integers such that $a+b+c+d = 45$ and no two of the integers differ by a multiple of 5.

8. A fair ten-sided die with the digits 0 to 9 (inclusive) on its faces is rolled three times. If M is the expected value of the largest possible three-digit number that can be formed using each of the three digits rolled exactly once, compute $\lfloor M \rfloor$, the greatest integer less than or equal to M. For example, if 4, 5, and 4 were rolled, the largest possible three-digit number would be 544. If three zeroes are rolled, the largest possible number is zero.

9. Compute the sum of all solutions to $\left(\log_a (2a-7)\right)\left(\log_{a-2} a\right) = -\dfrac{1}{2}$.

10. Let $\{x\}$ denote the fractional part of x. Compute the largest x such that $\{x\}^2 = x^2 - 56$.